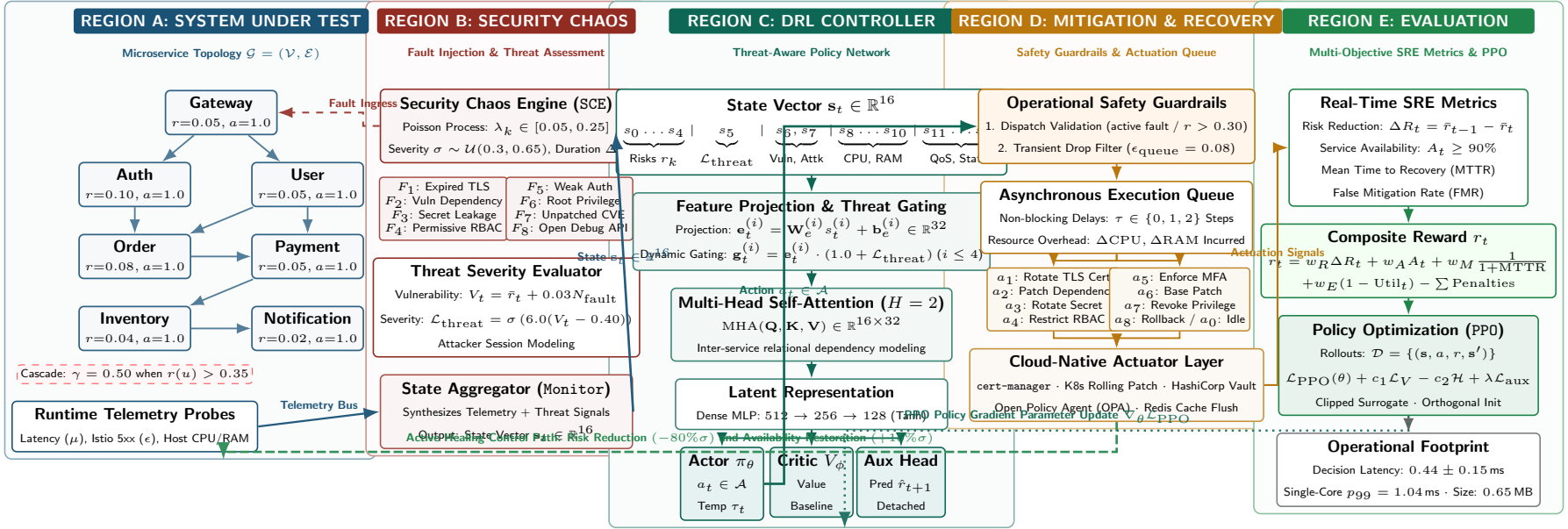

REGION A: SYSTEM UNDER TEST
Microservice Topology $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
Gateway
r=0.05, a=1.0
Auth
r=0.10, a=1.0
User
r=0.05, a=1.0
Order
r=0.08, a=1.0
Payment
r=0.05, a=1.0
Inventory
r=0.04, a=1.0
Notification
r=0.02, a=1.0
Cascade: $\gamma = 0.50$ when $r(u) > 0.35$
Runtime Telemetry Probes
Latency (μ), Istio 5xx (ϵ), Host CPU/RAM
Telemetry Bus
REGION B: SECURITY CHAOS
Fault Injection & Threat Assessment
Fault Ingress
Security Chaos Engine (SCE)
Poisson Process: $\lambda_k \in [0.05, 0.25]$
Severity $\sigma \sim \mathcal{U}(0.3, 0.65)$, Duration Δ
F_1: Expired TLS
F_2: Vuln Dependency
F_3: Secret Leakage
F_4: Permissive RBAC
F_5: Weak Auth
F_6: Root Privilege
F_7: Unpatched CVE
F_8: Open Debug API
Threat Severity Evaluator
Vulnerability: $V_t = \bar{r}_t + 0.03 N_{\text{fault}}$
Severity: $\mathcal{L}_{\text{threat}} = \sigma\,(6.0(V_t - 0.40))$
Attacker Session Modeling
State Aggregator (Monitor)
Synthesizes Telemetry + Threat Signals
Output: State Vector $\mathbf{s}_t \in \mathbb{R}^{16}$
REGION C: DRL CONTROLLER
Threat-Aware Policy Network
State Vector $\mathbf{s}_t \in \mathbb{R}^{16}$
$s_0 \cdots s_4$ | s_5 | s_6, s_7 | $s_8 \cdots s_{10}$ | $s_{11} \cdots$
Risks r_k
$\mathcal{L}_{\text{threat}}$
Vuln, Attk
CPU, RAM
QoS, Sta
Feature Projection & Threat Gating
Projection: $\mathbf{e}_t^{(i)} = \mathbf{W}_e^{(i)} s_t^{(i)} + \mathbf{b}_e^{(i)} \in \mathbb{R}^{32}$
Dynamic Gating: $\mathbf{g}_t^{(i)} = \mathbf{e}_t^{(i)} \cdot (1.0 + \mathcal{L}_{\text{threat}})\,(i \le 4)$
State $\mathbf{e}_t \in \mathbb{R}^{16}$
Action $a_t \in \mathcal{A}$
Multi-Head Self-Attention ($H = 2$)
$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \in \mathbb{R}^{16 \times 32}$
Inter-service relational dependency modeling
Latent Representation
Dense MLP: $512 \to 256 \to 128$ (Tanh)
Actor π_θ
$a_t \in \mathcal{A}$
Temp τ_t
Critic V_ϕ
Value
Baseline
Aux Head
Pred $\hat{r}_{t+1}$
Detached
REGION D: MITIGATION & RECOVERY
Safety Guardrails & Actuation Queue
Operational Safety Guardrails
1. Dispatch Validation (active fault / $r > 0.30$)
2. Transient Drop Filter ($\epsilon_{\text{queue}} = 0.08$)
Asynchronous Execution Queue
Non-blocking Delays: $\tau \in \{0, 1, 2\}$ Steps
Resource Overhead: ΔCPU, ΔRAM Incurred
a_1: Rotate TLS Cert
a_2: Patch Dependenc
a_3: Rotate Secret
a_4: Restrict RBAC
a_5: Enforce MFA
a_6: Base Patch
a_7: Revoke Privilege
a_8: Rollback / a_0: Idle
Cloud-Native Actuator Layer
cert-manager · K8s Rolling Patch · HashiCorp Vault
Open Policy Agent (OPA) · Redis Cache Flush
Actuation Signals
PPO Policy Gradient Parameter Update $\nabla_\theta \mathcal{L}_{\text{PPO}}$
Active Healing Control Path: Risk Reduction ($-80\%\sigma$) and Availability Restoration ($+1\%\sigma$)
REGION E: EVALUATION
Multi-Objective SRE Metrics & PPO
Real-Time SRE Metrics
Risk Reduction: $\Delta R_t = \bar{r}_{t-1} - \bar{r}_t$
Service Availability: $A_t \geq 90\%$
Mean Time to Recovery (MTTR)
False Mitigation Rate (FMR)
Composite Reward r_t
$r_t = w_R \Delta R_t + w_A A_t + w_M \frac{1}{1+\text{MTTR}}$
$+ w_E(1 - \text{Util}_t) - \sum \text{Penalties}$
Policy Optimization (PPO)
Rollouts: $\mathcal{D} = \{(\mathbf{s}, a, r, \mathbf{s}')\}$
$\mathcal{L}_{\text{PPO}}(\theta) + c_1 \mathcal{L}_V - c_2 \mathcal{H} + \lambda \mathcal{L}_{\text{aux}}$
Clipped Surrogate · Orthogonal Init
Operational Footprint
Decision Latency: 0.44 ± 0.15 ms
Single-Core $p_{99} = 1.04$ ms · Size: 0.65 MB